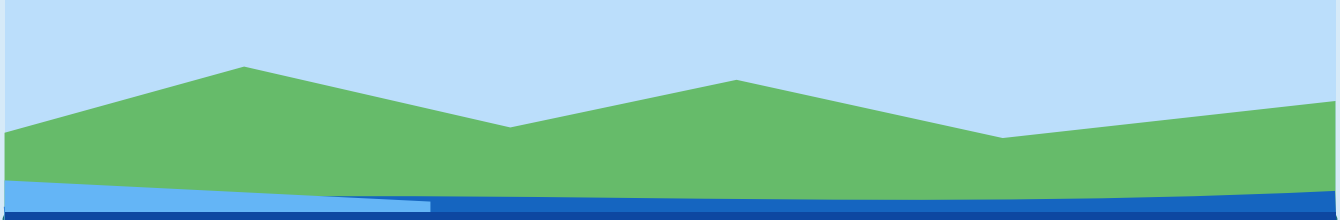
RIVERS

Hydrology · Processes · Landforms · Flooding · Management

RIVER EDEN (MEDC) | STORM DESMOND 2015 | NILE DELTA, EGYPT | HARD & SOFT ENGINEERING



River Eden, Cumbria · Storm Desmond December 2015 · Replace with real photograph for print

- ★ Explain the hydrological cycle and drainage basin processes
- ★ Describe river erosion, transportation and deposition
- ★ Explain the formation of waterfalls, V-valleys, meanders, oxbow lakes, floodplains, levees and deltas
- ★ Explain causes and impacts of flooding — Case Study: River Eden / Storm Desmond (2015)
- ★ Describe opportunities on floodplains and deltas — Case Study: Nile Delta, Egypt
- ★ Evaluate hard and soft engineering flood management strategies

■ SYLLABUS CHECKLIST — Tick when confident

Topic	Page
■ Explain the hydrological cycle and drainage basin characteristics	p.3
■ Describe river erosion, transportation and deposition	p.4
■ Explain formation of waterfalls, gorges, V-shaped valleys, interlocking spurs	p.5
■ Explain formation of meanders, oxbow lakes, floodplains, levees and deltas	p.6–7
■ Understand rivers present HAZARDS and OPPORTUNITIES	p.8–11
■ Explain physical and human causes of river flooding	p.8
■ Describe impacts — CASE STUDY: River Eden / Storm Desmond 2015	p.9
■ Evaluate hard and soft engineering flood management strategies	p.10
■ Describe opportunities — CASE STUDY: Nile Delta, Egypt	p.11

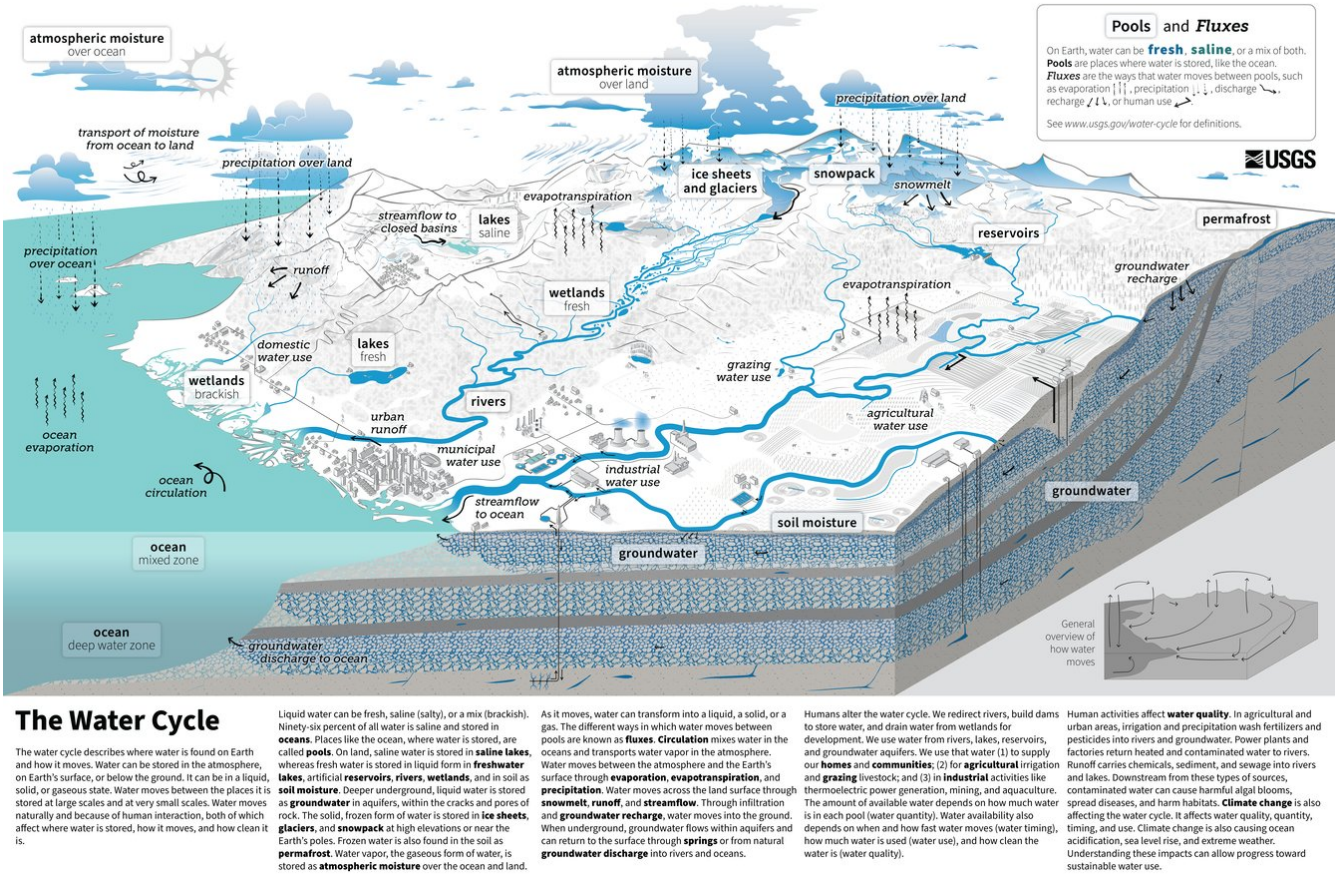
EXAM ALERT: Theme 2.2 → Paper 1, Question 3 or 4 (25 marks). Parts: 1+2+3+4+7 marks. The 7-mark question needs a named case study with specific statistics. Paper 2 skills: reading hydrographs, map work, % calculations.

■ KEY VOCABULARY

KEY TERM	DEFINITION
Drainage basin	Area of land drained by a main river and its tributaries
Watershed	Ridge of high land forming the boundary between two drainage basins
Discharge (Q)	Volume of water flowing past a point per second — measured in cumecs (m^3/s)
Hydraulic action	Force of water breaking away pieces of riverbank and bed
Abrasion	Sediment carried by river scrapes and erodes the channel walls and bed
Attrition	Sediment particles collide with each other, breaking into smaller rounder pieces
Solution	Soluble rocks (e.g. limestone) dissolve in the river water
Traction	Large boulders rolled along the riverbed by the river's flow
Saltation	Pebbles bounce along the riverbed in a leaping motion
Suspension	Fine particles of silt and clay carried within the river's flow
Meander	A bend in a river caused by differential erosion and deposition
Oxbow lake	Crescent-shaped lake formed when a meander neck is cut off during a flood
Floodplain	Flat, fertile land beside a river built up from layers of deposited alluvium
Levee	Natural raised bank alongside a river formed by deposition during flooding
Delta	Landform at a river's mouth where sediment is deposited as the river slows
Lag time	Time delay between peak rainfall and peak discharge on a hydrograph
Hard engineering	Building artificial structures to control a river (e.g. dams, embankments)
Soft engineering	Working with natural processes to reduce flood risk (e.g. afforestation, zoning)

■ PART A: THE HYDROLOGICAL CYCLE

The hydrological cycle is the continuous movement of water between the land, sea and atmosphere. It is a **CLOSED SYSTEM** — no water enters or leaves, it simply changes state and location.



Source: USGS Water Science School — Public Domain 2022

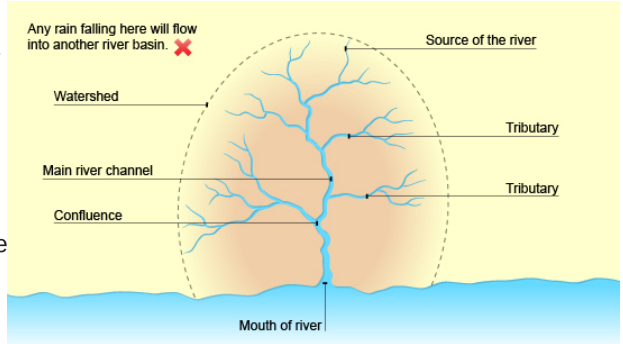
■ KEY TRANSFERS IN THE HYDROLOGICAL CYCLE

Transfer	Explanation
Precipitation	Water falling as rain, snow, sleet or hail from atmosphere to surface
Evapotranspiration	Evaporation from water surfaces PLUS transpiration from plants — water returns to atmosphere
Interception	Water caught on leaves and branches before reaching the ground
Infiltration	Water soaking downward from the surface into the soil
Throughflow	Water moving laterally (sideways) through the soil layer towards the river
Percolation	Water moving downward through soil into the bedrock/groundwater
Overland flow	Water flowing over the surface when ground is saturated or impermeable — fastest route to river
Groundwater flow	Very slow movement of water through permeable rock towards rivers and sea

■ THE DRAINAGE BASIN SYSTEM

A **drainage basin** is an area of land drained by one main river and all its **tributaries**. The boundary is called the **watershed** — a ridge of high land. Drainage basins are **OPEN SYSTEMS** — water enters as precipitation and leaves via evapotranspiration and river flow to the sea.

Key terms: Source (start of river) · Mouth (where river meets sea) · Tributary (smaller river joining main channel) · Confluence (where two rivers meet) · Watershed (boundary of drainage basin)



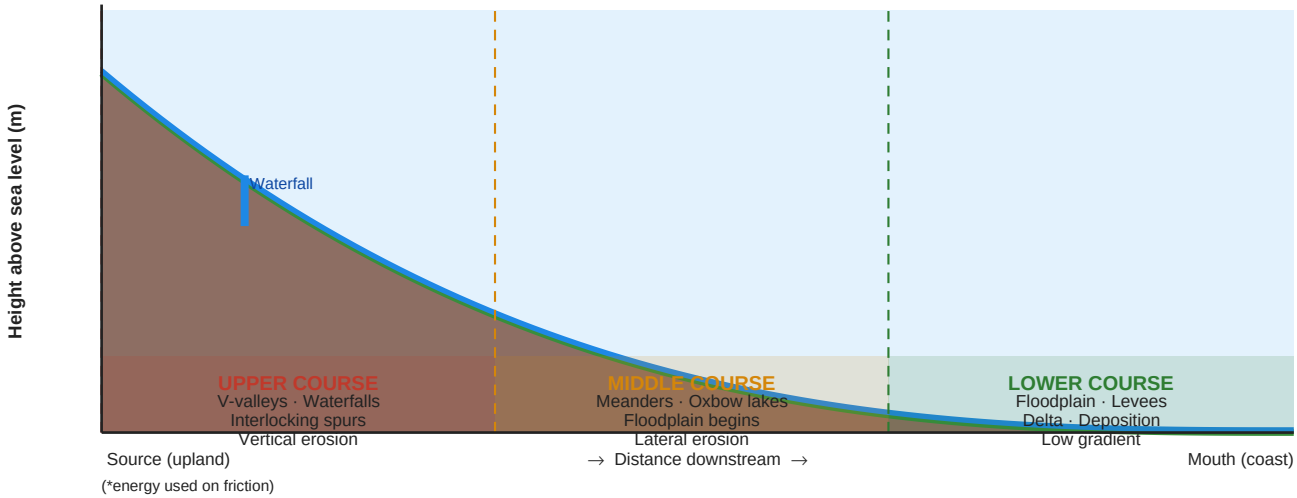
Drainage basin system showing watershed, tributaries, confluence, source and mouth

■ PART A: RIVER PROCESSES — EROSION, TRANSPORT & DEPOSITION

These three processes operate throughout the river's course and create ALL river landforms. You must be able to name and explain each one.

EROSION	TRANSPORTATION	DEPOSITION
■ Hydraulic action — force of water breaks off bank/bed material	■ Traction — large boulders rolled along riverbed	■ Occurs when river LOSES ENERGY and slows down
■ Abrasion — sediment scrapes and wears away channel	■ Saltation — pebbles bounce along riverbed	■ On inside of bends (slower water)
■ Attrition — particles collide, get smaller and rounder	■ Suspension — fine particles carried within the flow	■ Where river meets sea (delta formation)
■ Solution — soluble rock dissolves in river water	■ Solution — dissolved material invisible in water	■ After a flood — load dropped on floodplain
→ Greatest in UPPER COURSE	→ All four occur throughout	→ Heaviest material first

■ THE RIVER LONG PROFILE — How the river changes from source to mouth



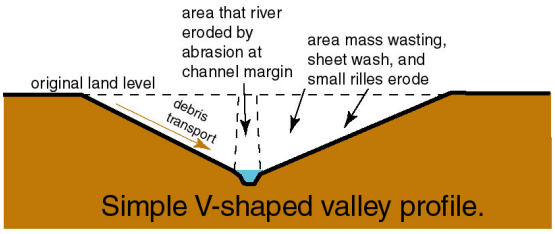
***Note on velocity:** Although friction is high in the upper course, velocity actually increases downstream as the channel becomes wider and deeper and less water is in contact with the bed relative to the total volume.

■ PART B: UPPER COURSE LANDFORMS — V-Shaped Valleys & Waterfalls

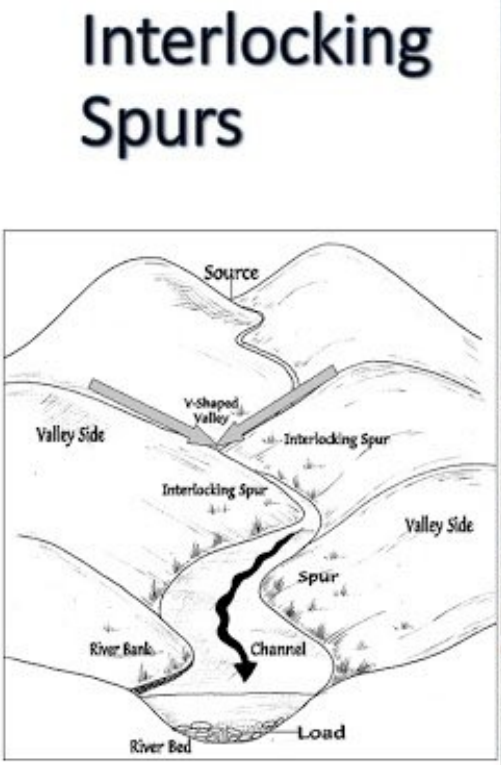
V-Shaped Valleys & Interlocking Spurs

1. In the upper course, the river has a **steep gradient** and cuts **DOWNWARDS** (vertical erosion) using hydraulic action and abrasion
2. The valley sides are left steep and exposed to weathering and mass movement (soil creep, slumping, freeze-thaw)
3. Weathered material slides down into the river channel and is carried away downstream
4. The result is a **narrow, steep-sided V-shaped valley**
5. Where the river meets hard resistant rock, it flows around it — leaving **interlocking spurs** projecting from alternate sides like the teeth of a zip
6. From above, these spurs **interlock** — the river winds between them

Also in upper course: Potholes — cylindrical holes drilled into the riverbed when stones caught in a hollow spin around by eddying water, grinding the rock down by abrasion.

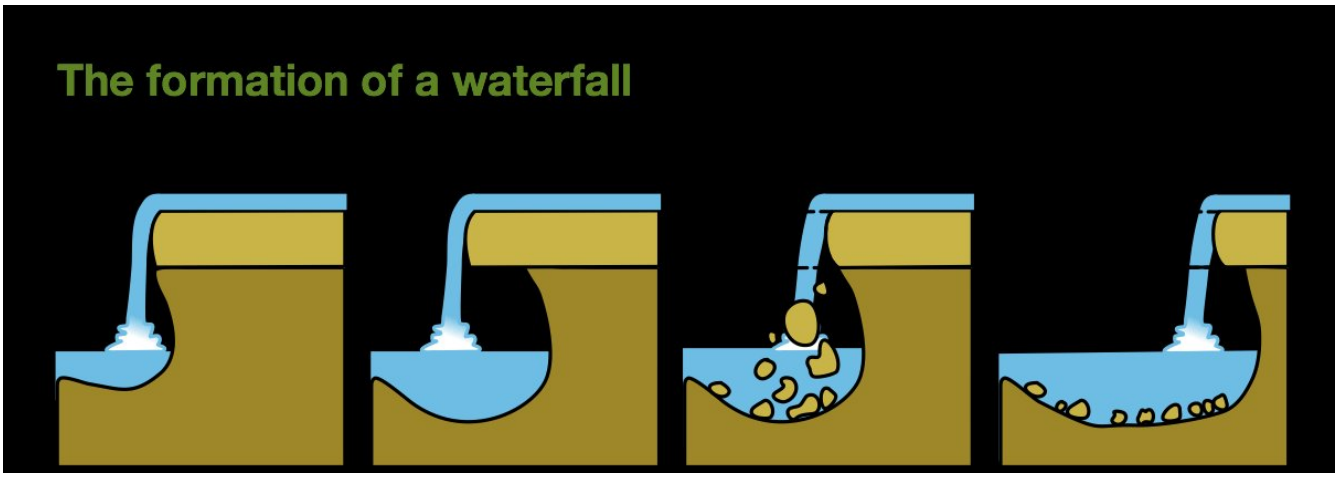


V-shaped valley cross-section showing steep sides, narrow floor and river at base



Interlocking spurs — diagram (left) showing V-shaped valley, river channel and projecting spurs; photograph (right) of classic upland valley with interlocking spur landform.

■ WATERFALL & GORGE FORMATION



Formation of a waterfall — 4 stages showing undercutting, plunge pool development and upstream retreat leaving a gorge. Source: internetgeography.net

High Force Waterfall, River Tees, County Durham — hard whinstone overlies softer limestone and shale. One of England's highest waterfalls at 21m. Classic IGCSE case study example.

■ Paper 1 · Q3(b) [3 marks]

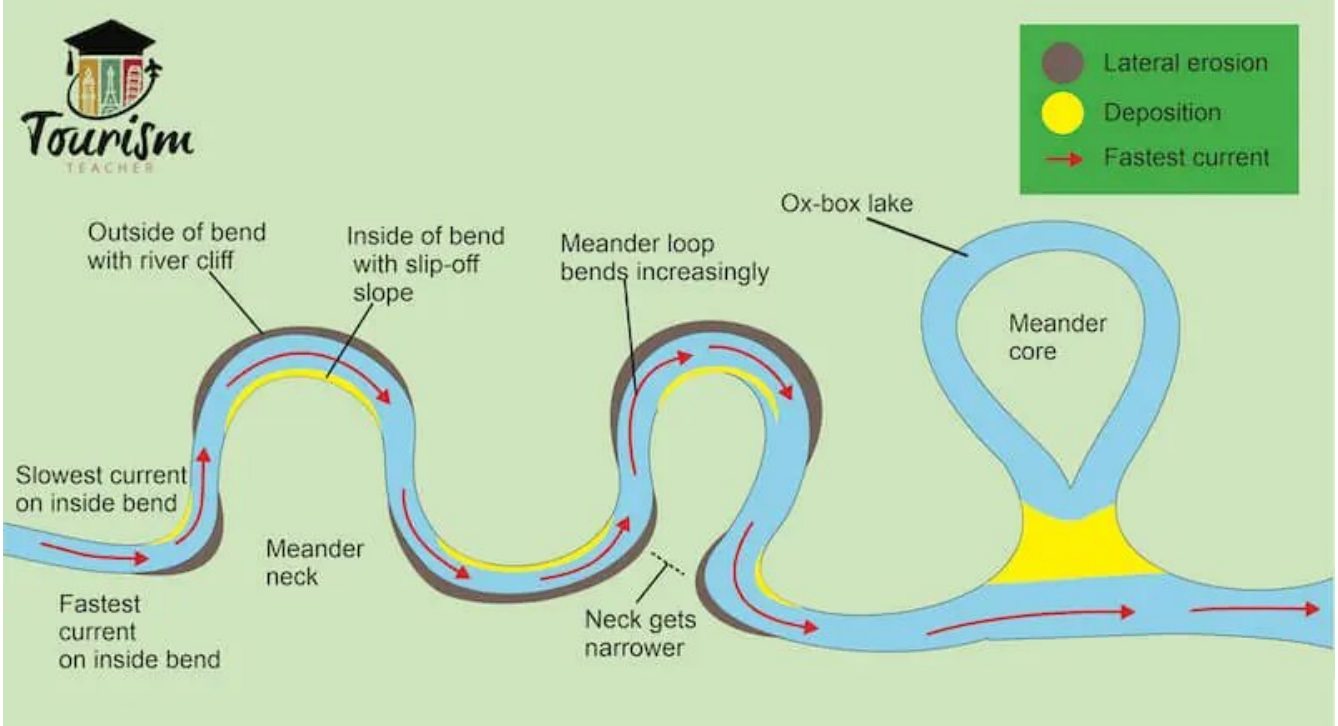
Explain how a waterfall is formed. [3 marks]

✓ Mark Scheme:

Hard rock overlies soft rock [1]; soft rock eroded faster by hydraulic action/abrasion creating undercutting/notch [1]; hard rock overhang collapses into plunge pool [1]; waterfall retreats upstream leaving steep-sided gorge [1]. Max 3.

■ PART B: MIDDLE COURSE — Meanders & Oxbow Lakes

In the middle course the river has more lateral (sideways) energy. Differential velocity around bends drives both erosion and deposition.



Meander and oxbow lake formation — outer bank erosion (river cliff), inner bank deposition (slip-off slope), neck narrowing, flood cutoff and oxbow lake. Source: tourismteacher.com

Feature	Explanation
Outer bank (river cliff)	Water flows FASTEST — hydraulic action and abrasion undercut the bank, creating steep river cliff
Inner bank (slip-off slope)	Water flows SLOWEST — deposition of sand and gravel builds up a gently sloping point bar
Helical flow	Corkscrew-shaped water movement that transfers eroded material from outer to inner bank
Neck	Narrow strip of land between two loops of the meander — cut through during a flood
Oxbow lake	Isolated crescent lake when deposition seals off old meander loop at both ends

River Cuckmere, East Sussex — classic IGCSE photograph of meanders on a floodplain. S-shaped bends, outer river cliffs and inner slip-off slopes clearly visible.

■ Paper 1 · Q3(c) [4 marks]

Describe and explain how an oxbow lake is formed. [4 marks]

✓ Mark Scheme:

River erodes outer bank of meander by hydraulic action/abrasion [1]; meander tightens — neck of land narrows [1]; during a flood, river cuts through the neck taking shortest route [1]; deposition seals off old meander loop at both ends forming crescent oxbow lake [1].

■ PART B: LOWER COURSE — Floodplain, Levees & Deltas

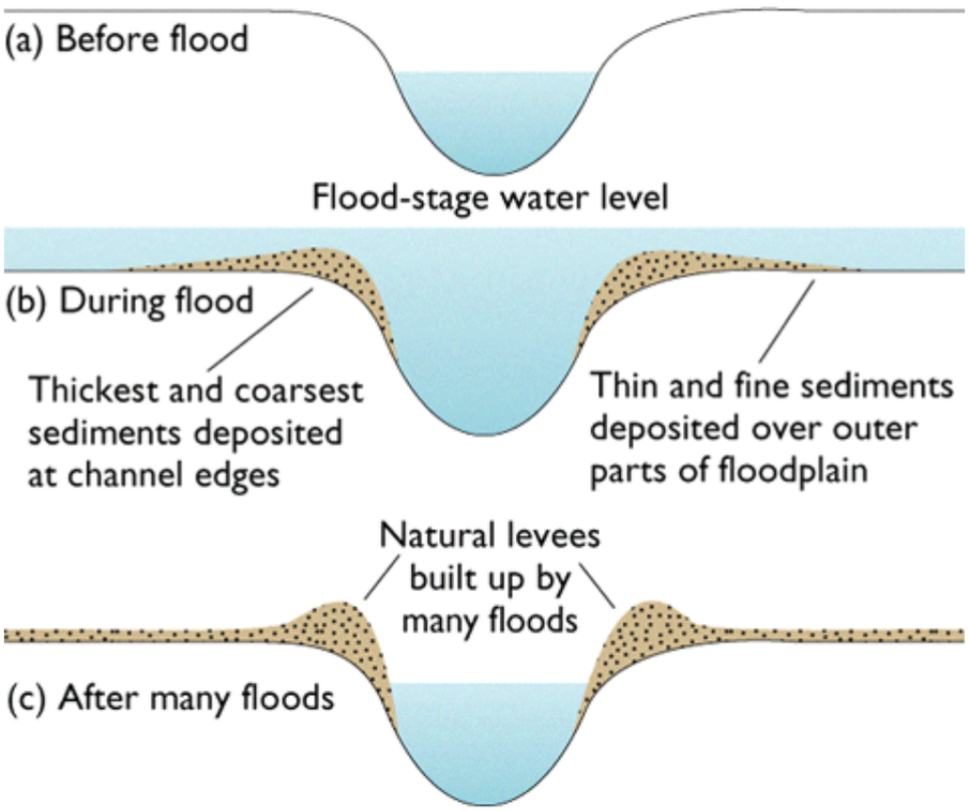
Floodplain Formation

1. In the lower course, river has low gradient and high volume
2. During floods, water spills over banks onto flat land beside the river
3. Velocity drops instantly → deposition of **alluvium** (fertile silt)
4. Repeated flooding builds up flat, fertile floodplain over time
5. Meanders migrate sideways across the valley, widening it further

Levee Formation

1. When river floods, it immediately slows as it leaves the channel
2. Heaviest material deposited FIRST — right beside the channel banks
3. Builds up raised natural embankments (levees) alongside the river
4. Each flood adds another layer — levees grow taller over time
5. Lighter silt deposited further from the channel, across the floodplain

Example: Mississippi River levees reach up to 15m high. River Nile levees built up the delta over thousands of years.



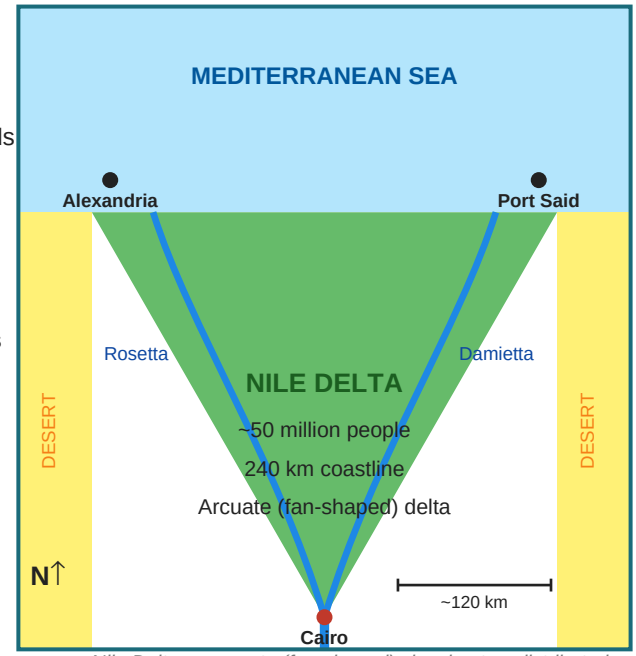
Levee cross-section showing coarse deposition near channel and fine silt on floodplain

■ DELTA FORMATION

1. River reaches its **mouth** — meets the sea or lake
2. Velocity drops to almost zero → **all remaining load deposited**
3. If deposition is faster than removal by waves, land builds up
4. River splits into smaller channels called **distributaries**
5. Fan-shaped landform grows outward into the sea over time

3 Types of Delta: Arcuate (fan-shaped) e.g. Nile · **Bird's foot** e.g. Mississippi · **Cuspate** (pointed) e.g. Ebro

Conditions needed: large sediment load · weak currents/tides · shallow coastal water · regular deposition exceeding removal



Nile Delta — arcuate (fan-shaped) showing two distributaries

■ PART C: RIVER FLOODING — Causes & The Hydrograph

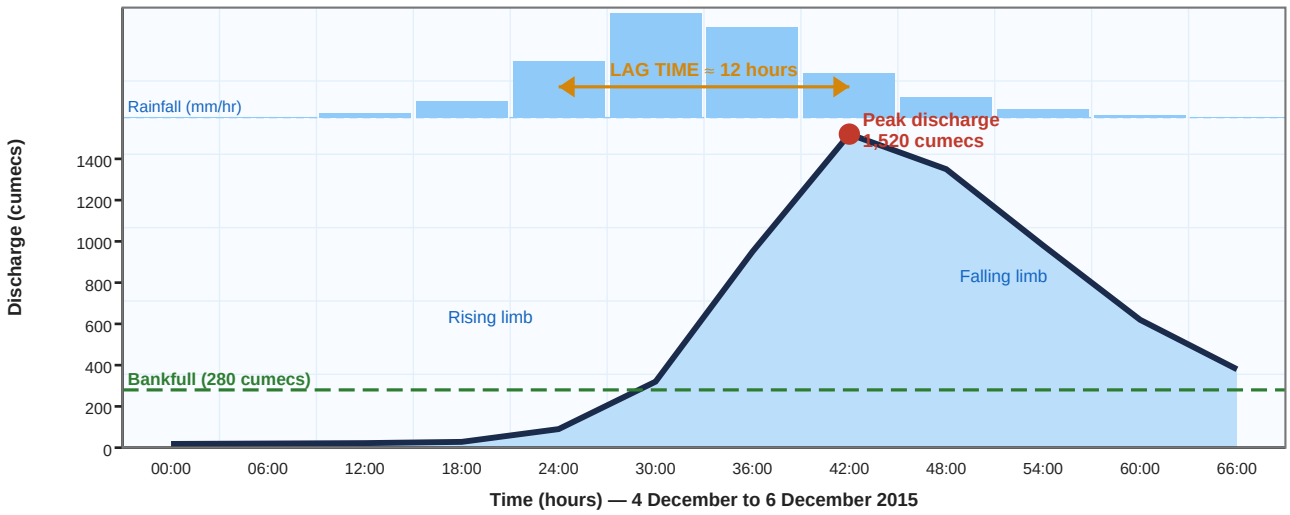
A river **floods** when its discharge exceeds its **bankfull capacity** and water spills over the banks onto the floodplain.

■ Physical Cause	Why it causes flooding
Prolonged/heavy rainfall	Ground saturates — overland flow increases rapidly
Snowmelt	Large volumes of meltwater added quickly to rivers
Steep relief	Water runs off slopes fast — very short lag time
Impermeable rock/clay	Water cannot infiltrate — runs straight to river
Low vegetation	Less interception — more water reaches river faster
Orographic rainfall	Moist air rises over hills — intense rain (e.g. Lake District)

■ Human Cause	Why it causes flooding
Urbanisation	Tarmac/concrete impermeable — drains deliver water rapidly
Deforestation	Fewer trees = less interception/transpiration → more runoff
Building on floodplain	Reduces natural flood storage capacity
Channel straightening	Speeds water through → increases flooding downstream
Climate change	More intense storms — atmospheric rivers (e.g. Storm Desmond)

■ THE STORM HYDROGRAPH

Storm Hydrograph — River Eden, Storm Desmond December 2015



PAPER 2 TIP — Reading a hydrograph: Always quote specific data (e.g. 'peak discharge of 1,520 cumecs at hour 36'). Name features: rising limb / falling limb / lag time / bankfull level. A FLASHY hydrograph = short lag time + high peak = urban / impermeable / steep. A SUBDUED hydrograph = long lag time + low peak = rural / forested / permeable.

■ Paper 2 Skills [3 marks]

Study the hydrograph. Describe how discharge changed during Storm Desmond. Use data in your answer. [3 marks]

✓ Mark Scheme:

Discharge rose rapidly from ~20 cumecs to peak of 1,520 cumecs at hour 36 [1]; lag time between peak rainfall and peak discharge was approximately 12 hours [1]; after the peak, discharge fell more slowly over the following 30 hours (falling limb) [1]; discharge exceeded bankfull level (280 cumecs) for approximately 18 hours [1]. Max 3 — data must be quoted.

CASE STUDY: RIVER EDEN, CUMBRIA — STORM DESMOND, DECEMBER 2015

Location: NW England, drains Lake District fells to Solway Firth · Catchment: 2,275 km² · Main city: Carlisle (pop. 74,000) · Previous floods: 2005, 2009, 2015

Storm Desmond flooding, Cumbria, December 2015 — aerial view of flooded residential streets. 5,200 properties flooded across Cumbria.

Physical Causes

- **Storm Desmond** — deep Atlantic low pressure, stalled over Cumbria 4–5 Dec 2015
- **341.4mm of rain in 24 hours** at Honister Pass — new UK record (previous: 316mm in 2009)
- **Orographic rainfall** — moist air forced up over Lake District fells, condensing as intense rain
- **Already saturated ground** — November 2015 was 2nd wettest November since 1910; zero infiltration capacity left
- **Radial drainage pattern** — rivers drain outwards from the dome of the Lake District, concentrating flow rapidly
- **Atmospheric river** — unusually moist airstream from the Caribbean linked to climate change

Human Causes

- **Urbanisation** in Carlisle — impermeable surfaces increase overland flow into River Eden
- **Deforestation** in upper Eden catchment — reduced interception and shorter lag time
- **Defences overtopped** — £38m Carlisle scheme (post-2005) designed for 1-in-200-year event; 2015 exceeded this
- **Building on floodplain** — new developments reduced natural flood storage capacity
- **Climate change** — warmer Atlantic produces more intense atmospheric rivers and rainfall events

■ CASE STUDY: RIVER EDEN — Impacts of December 2015 Flooding

Social Impacts	Economic Impacts	Environmental Impacts
■ 5,200 properties flooded across Cumbria	■ 3,000 businesses affected; farmland destroyed	■ Thousands of riverside trees uprooted
■ 2 fatalities (including man swept into River Kent, Kendal)	■ West Coast Main Line (London–Glasgow) severed for days	■ Landslides across Lake District hillsides
■ 60,000 homes without electricity at peak	■ 3 bridges swept away — 14 more damaged	■ Floodplain soils stripped of fertile topsoil
■ 45 schools closed — 3,034 children affected	■ 354.8 km of carriageway damaged	■ Farmland contaminated with sediment and waste
■ 1,051 medical appointments cancelled/rescheduled	■ Government pledged £50 million repair and renew grant	■ River channels significantly altered by sediment
■ Thousands of people evacuated from their homes	■ Total repair cost estimated at £500 million+	■ Habitats along River Eden corridor damaged
■ Pooley Bridge (built 1764) swept away — community cut off	■ Tourism income lost across Lake District	■ Waterfalls reactivated in unusual locations

■ Paper 1 · Q3(d) [4 marks]

Describe the social and economic impacts of flooding for a named river. [4 marks]

✓ Mark Scheme:

Social: 5,200 properties flooded in Cumbria [1]; 60,000 homes without electricity [1]; 2 deaths / 45 schools closed [1]. Economic: 3 bridges destroyed including Pooley Bridge [1]; West Coast Main Line cut [1]; £50m government repair scheme [1]. Max 4 — must name River Eden/Cumbria.

■ PART D: MANAGING FLOOD IMPACTS — Hard & Soft Engineering

Strategy	Description	Cumbria Example	■ Advantage	■ Disadvantage
Flood embankments / levees	Raised concrete/earth banks alongside river	£38m Carlisle scheme on Eden, Petteril & Caldew	Protects many properties; reliable	Expensive; can increase flooding downstream
Dams & reservoirs	Control discharge by storing water upstream	Thirlmere reservoir in Eden catchment	Regulate flow; provide water supply	Very expensive; floods land; disrupts ecology
Channel straightening	Remove bends to speed water through area	Used on Cumbrian tributaries	Quick to build; reduces local risk	Increases flooding downstream; damages habitats
Afforestation	Plant trees in catchment to intercept rainfall	'Slow the Flow' tree planting in Eden headwaters	Sustainable; cheap; improves habitat	Takes years to mature; needs large areas
Floodplain zoning	Restrict development on natural floodplains	Planning restrictions in Carlisle floodplain	Cheap; prevents future damage; sustainable	Restricts development; hard to enforce retroactively
Flood warning systems	Sensors + EA alerts give communities time to prepare	EA FloodLine; gauges on Eden at Sheepmount	Saves lives; cheap; community resilience	Does not prevent flooding; relies on public action
Washlands	Allow designated land to flood naturally upstream	Floodplain areas left as washlands above Carlisle	Natural; maintains floodplain ecology; cheap	Land sacrificed; farmers need compensation

■ CUMBRIA'S RESPONSE AFTER DECEMBER 2015

- Government invested **£72 million** to protect 4,300 more homes by 2021
- Cumbria Floods Partnership formed; **Cumbria Flood Action Plan** published 2016
- '**Slow the Flow**' catchment management — tree planting and leaky dams on upland streams
- 7 new flood warning areas created; improved gauging network on River Eden
- New flood protection schemes at Appleby, Keswick, Threlkeld (Gate Gill Beck)
- Carlisle defences reviewed and raised to withstand higher return-period events

7-MARK PEEL FRAMEWORK

P — POINT	Name a strategy or feature with a located example
E — EVIDENCE	Quote a specific statistic from your case study (341.4mm, 5,200 homes, £72m, 50 million people)
E — EXPLAIN	Explain the process / cause / impact using geographical terminology
L — LINK / EVALUATE	Evaluate: how successful? Limitations? (e.g. defences were overtopped in 2015)

■ Paper 1 · Q3(f) [7 marks]

For a named river you have studied, describe the impacts of flooding and evaluate the strategies used to manage them. [7 marks]

✓ Mark Scheme:

Level 3 (6–7): Named river + specific data throughout + balanced impacts (social/economic/environmental) + evaluation of management (e.g. hard vs soft, success/limitations) + concluding judgement. Level 2 (4–5): Some data, describes impacts and management. Level 1 (1–3): Basic impacts listed. KEY: name River Eden + statistics (341.4mm, 5,200 homes, £72m) + evaluate hard vs soft.

CASE STUDY: NILE DELTA, EGYPT — Opportunities of Living on a Floodplain/Delta

Location: Northern Egypt — Nile meets Mediterranean Sea · ~50 million people (40% of Egypt) · Population density >1,000/km² · 240 km Mediterranean coastline

■ Agriculture

■ **Fertile alluvial soil** — annual Nile floods deposited nutrient-rich silt for 5,000 years, creating some of the world's most productive farmland

■ **60% of Egypt's population** depends on agriculture as their main income source

■ Key crops: **wheat, maize, rice, cotton** (major export), citrus fruits, vegetables

■ Population density >1,000/km² — among world's most densely farmed areas

■ Water Supply

■ The Nile provides **95% of Egypt's freshwater** — critical in a desert nation

■ Irrigation canals criss-cross the delta, supporting year-round cultivation

■ **Lake Nasser** (Aswan High Dam) stores water for dry-season irrigation

■ Settlement, Industry & Trade

■ Major cities: **Cairo** (20m+), Alexandria — Egypt's economic heartland

■ **Suez Canal** at eastern edge — one of world's most important shipping routes

■ Fishing industry in delta lagoons and Mediterranean coast

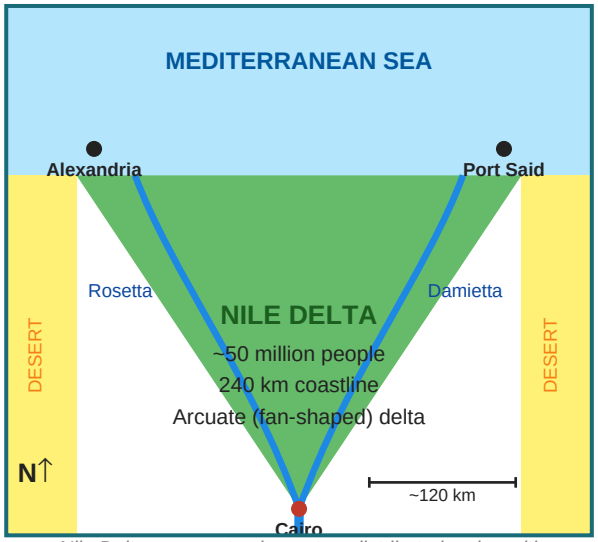
■■ Challenges & Threats

■ **Aswan High Dam (1970)** — stopped annual floods; silt trapped in Lake Nasser; soils now need expensive fertilisers

■ **Sea level rise** — delta only 1m above sea level; 25% could be submerged by 2100

■ **74,600 hectares** of farmland lost to urban expansion 1992–2015

■ **Saltwater intrusion** — reducing coastal crop yields



Nile Delta — arcuate shape, two distributaries, key cities

Key Statistics

Population	~50 million (40% of Egypt)
Pop. density	>1,000 per km²
Coastline	240 km
Farmland lost	74,600 ha (1992–2015)
Sea level risk	887,000 people at risk from 1m rise
Nile allocation	55.5 billion m³ per year

■ Paper 1 · Q3(d) [4 marks]

Describe the opportunities that living on a river delta provides for people. Refer to a named example. [4 marks]

✓ Mark Scheme:

1 mark per valid opportunity + development. Fertile alluvial soil enables intensive agriculture [1] — Nile Delta produces wheat, rice, cotton for 50m residents [1]; Nile provides 95% of Egypt's freshwater for irrigation [1]; Suez Canal enables international trade [1]; major cities Cairo/Alexandria [1]. Must name Nile Delta. Max 4.

■ QUICK-FIRE REVISION — 10 Questions (30 seconds each)

No.	Question	Your Answer
1	Name the FOUR types of river erosion.	
2	What is the difference between infiltration and throughflow?	
3	Define lag time. Name TWO factors that shorten it.	
4	Explain what happens on the OUTER BANK of a meander.	
5	Give TWO physical causes of the 2015 Cumbria floods. Include a statistic.	
6	State THREE social impacts of Storm Desmond with statistics.	
7	Difference between hard and soft engineering? One example each.	
8	Why is the Nile Delta so densely populated? Give TWO reasons.	
9	How did the Aswan High Dam change the Nile Delta? One positive + one negative.	
10	Name the four types of river transport. Which carries the largest material?	

■ PAPER 2 DATA SKILLS — River Eden Discharge Data

Study Table A below showing discharge data for four rivers during Storm Desmond, December 2015.

River	Peak discharge (cumecs)	Lag time (hrs)	Catchment (km²)	Land use
Eden at Sheepmount	1,520	12	2,275	Mixed upland fell + urban (Carlisle)
Greta at Keswick	398	6	184	Mostly upland fell, some urban
Petteril at Harraby	156	18	446	Mixed agricultural / suburban
Kent at Burneside	245	9	209	Upland fell + small towns

■ P2(a)(i) [1 mark]

Using Table A, identify the river with the highest peak discharge.

_____

_____

✓ Mark Scheme:

River Eden at Sheepmount [1].

■ P2(a)(ii) [2 marks]

Calculate the difference in lag time between the River Greta and the River Petteril. Show your working.

✓ Mark Scheme:

18 – 6 = 12 hours [2]; correct working but arithmetic error = [1].

■ P2(a)(iii) [3 marks]

Using Table A, suggest reasons why the River Greta has a shorter lag time than the River Petteril. [3 marks]

✓ Mark Scheme:

Greta has smaller catchment (184 vs 446 km²) so water reaches river faster [1]; drains steep upland fell with thin soils and impermeable rock — rapid overland flow [1]; Petteril drains more agricultural land with greater infiltration capacity, lengthening lag time [1]. Max 3.

■ EXAM MARK BAND GUIDANCE — Theme 2.2 Rivers

Marks	What the examiner expects	Common errors
1 mark	State one correct fact, term or process	Too vague: 'the river erodes' — which type?
2–3 marks	Describe with explanation; at least one named example or statistic	No data; no place name; just listing
4 marks	Two fully developed points with explanation + located example	Two points but only one developed; no statistics
7 marks	Named case study throughout; specific data; balanced social/economic/environmental impacts; evaluate management strategies; evaluative conclusion	Generic answer; no statistics; no evaluation; ignoring the word 'evaluate'

■ REVISION SUMMARY CARD

PROCESSES & LANDFORMS	RIVER EDEN / CUMBRIA	NILE DELTA, EGYPT
Erosion: hydraulic action, abrasion, attrition, solution Transport: traction, saltation, suspension, solution Deposition: inner bends, mouth, after floods Upper: waterfall, gorge, V-valley, spurs Middle: meanders, oxbow lakes Lower: floodplain, levees, delta Hydrograph: lag time, peak Q, bankfull level	Storm Desmond 5-6 Dec 2015 • 341.4mm rain in 24hrs (UK record) • Saturated ground from wet Nov • Peak discharge: 1,520 cumecs IMPACTS: • 5,200 properties flooded • 60,000 homes without power • 3 bridges swept away • 2 deaths; 45 schools closed MANAGEMENT: • £72m investment post-2015 • 'Slow the Flow' afforestation • Improved flood warnings • New defences: Appleby, Keswick	Location: N. Egypt Nile meets Mediterranean • ~50m people; 40% of Egypt • Pop. density >1,000/km² • 240km coastline OPPORTUNITIES: • Fertile alluvium — wheat, rice • 95% of Egypt's freshwater • Suez Canal — global trade • Cairo (20m+), Alexandria CHALLENGES: • Aswan Dam stopped silt • Sea level rise — 25% delta • 74,600 ha farmland lost

Resource: www.geographypods.com/rivers · Syllabus: cambridgeinternational.org/0460 · Revision Padlet: SAIS Geography